PARAMETER-EFFICIENT FINE-TUNING

LoRA

& QLoRA

Teaching AI new tricks — without breaking the bank!

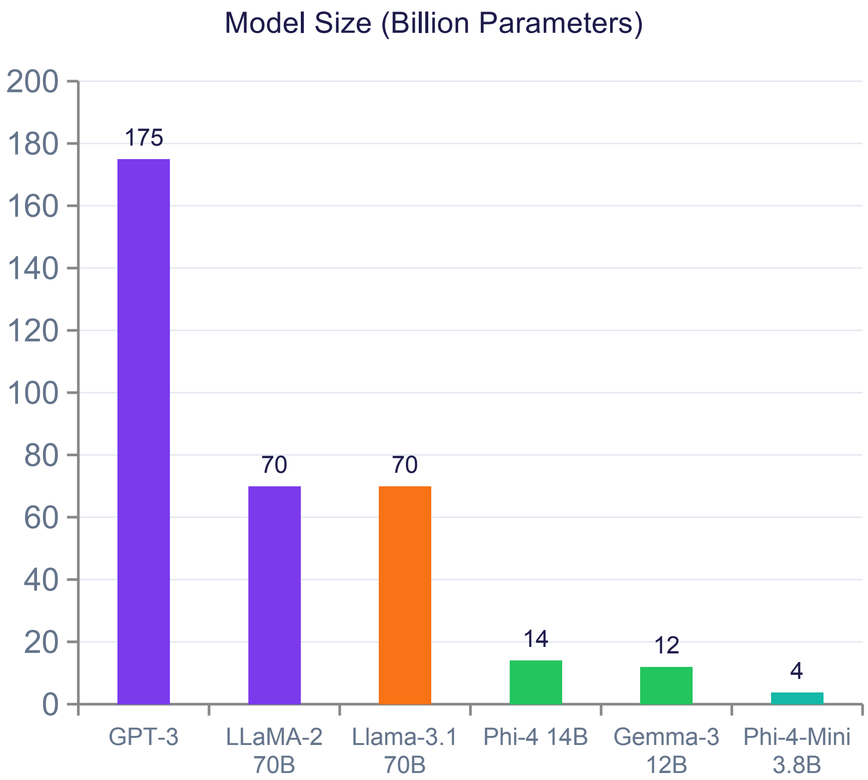
LLMs Are GIANT Brains!

How big is a language model?

- GPT-3 has 175 BILLION numbers (parameters)
- Llama 3.1 (405B) has 405 billion numbers
- Just storing GPT-3 takes 700 GB of space
- Training GPT-3 used 570 GB of internet text
- ChatGPT learned from more text than you could read in 100,000 years!

Simple English!

Imagine if you read EVERY book, website, article, and Wikipedia page ever written. That's what an LLM does!



Fine-Tuning & The Cost Problem

What is Fine-Tuning?

🧠 LLMs already know LOADS from pre-training

🎯 Fine-tuning = teach it ONE specific skill

💬 Examples: become a doctor AI, legal assistant, or coding helper

📚 You show it new examples, it learns from them

💡 **Simple English!**
You already know how to read & write. Fine-tuning is like taking a special cooking class — you gain new skills without forgetting how to read!

But... Full Fine-Tuning is EXPENSIVE!

🏎️ Train GPT-3 from scratch
$4,600,000 = ~46 Lamborghinis!

💸 Full fine-tune LLaMA-2 70B
$50,000+ = 8 A100 GPUs + weeks!

🎉 LoRA fine-tune Llama-3 7B
~$10 = One pizza dinner!

🚀 QLoRA fine-tune 70B model
~$50-100 = Single consumer GPU!

LoRA

LoRA = Magic Sticky Notes!

Low-Rank Adaptation of Large Language Models • Hu et al., Microsoft 2021

😰

Full Fine-Tuning

- Change ALL billions of weights
- Super expensive
- Needs huge GPUs
- Takes weeks

📝

LoRA

- Freeze the original brain
- Add tiny 'adapter' layers
- Only train the adapters!
- 100× cheaper & faster

🎉

The Result?

- Same smart brain
- New specialized skills
- Tiny file (~50MB!)
- Swap adapters freely!

How LoRA Works: The Maths Made Fun!

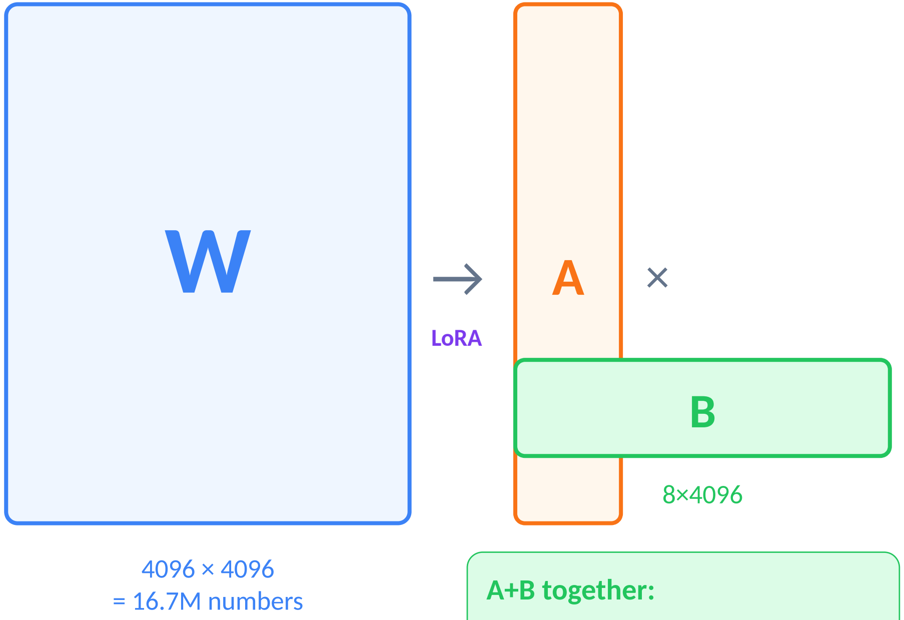
The Big Idea:

Instead of changing one GIANT matrix (W), LoRA learns two TINY matrices (A and B) whose product approximates the change!

$$W\_new = W + B \times A$$

W	The original frozen matrix (HUGE: 4096×4096)
A	Tiny matrix — tall & thin (e.g. 4096 × 8)
B	Tiny matrix — short & wide (e.g. 8 × 4096)
rank=8	The '8' is the rank — like the number of lanes on a highway!

💡 Bookmarks on a giant book — not rewriting every page!



A+B together:

65,536 numbers
= 256× FEWER than W!

LoRA's 3 Key Settings (The Control Knobs!)

🔢 Rank (r)

How many 'shortcuts' does LoRA use?

- r = 4 → Smallest, fastest (simple tasks)
- r = 8 → Most popular sweet spot
- r = 16 → Better quality, more memory
- r = 64 → Almost like full training!

💡 Most people start with r=8 or r=16. Higher rank = better results, but costs more!

📢 Alpha (α)

How LOUD should the new shortcuts be?

- alpha = 8 (with r=8) → 1× strength
- alpha = 16 (with r=8) → 2× strength
- alpha = 32 (with r=8) → 4× strength
- Rule of thumb: alpha = 2 × rank

💡 Alpha controls how much the LoRA adapters influence the output. Too high = forget original skills!

🎯 Target Modules

Which parts of the brain do we update?

- q_proj, v_proj → Attention (most popular)
- k_proj, o_proj → Also attention
- All linear layers → Best quality!
- lm_head → Output layer

💡 Attention layers (q,v) are usually enough. Add more layers for harder tasks!

LoRA in the Wild: Real Models 2021 – 2025

2021

LoRA Paper (Microsoft)
GPT-3, GPT-2 — 10,000× fewer params trained vs full!

First LoRA experiments
Matched full fine-tuning quality on NLP benchmarks

2022

Stable Diffusion LoRA
Artists create custom art styles in minutes for FREE!

SD character LoRAs
CivitAI launches — 100,000+ LoRA models today!

2023

Alpaca-LoRA
"The $100 ChatGPT" — LLaMA-7B fine-tuned for $100!

WizardLM
Smarter reasoning via LoRA on LLaMA — beat ChatGPT!

Mistral-7B LoRA ecosystem
Thousands of community fine-tunes appear overnight

Code Llama (Meta)
Coding specialist — LLaMA2 with LoRA for code

2024

Llama-3 8B / 70B (Meta)
Massive LoRA ecosystem; best open model in 2024

Phi-3 Mini / Small / Medium
Microsoft's tiny-but-smart models, LoRA-friendly

Flux.1 Dev / Schnell LoRAs
Revolutionized AI image generation — photorealistic!

Gemma 2 (Google)
2B, 9B, 27B — great quality with LoRA fine-tuning

2025

Llama 4 Scout 17B (Meta)
Mixture-of-Experts — LoRA fine-tunes for specialization

DeepSeek-R1 (China)
Reasoning champion — community LoRA fine-tunes everywhere

Gemma 3 (Google)
1B–27B family; 4B fits easily on laptop with LoRA!

Qwen3 (Alibaba)
Multilingual powerhouse with strong LoRA support

LoRA for Art! (Image Generation Models)

Did you know LoRA started a REVOLUTION in AI art? 🎨

2022

Stable Diffusion 1.5

First big LoRA wave! Artists trained custom characters & styles in hours on a home PC.

⚡ **Fun Fact:**

SD 1.5 LoRA = ~5MB file. Works on ANY GPU!

2023

SDXL LoRA

Higher resolution, more realistic. SDXL LoRAs enabled photorealistic portraits.

⚡ **Fun Fact:**

CivitAI now has 100,000+ LoRA models for download!

2024

Flux.1 Dev / Schnell

Game-changer! Photorealistic humans, consistent characters, stunning quality.

⚡ **Fun Fact:**

Flux.1 LoRAs created in under 20 minutes on one GPU!

2025

Wan2.1 (Video LoRAs)

LoRAs now work for VIDEO generation! Custom motion styles, characters in motion.

⚡ **Fun Fact:**

Train a video style LoRA — under $5 of compute!

What is QLoRA? (Even Smarter Than LoRA!)

QLoRA = Quantized LoRA • Dettmers, Pagnoni, Holtzman et al. — "QLoRA: Efficient Finetuning of Quantized LLMs" (2023)

The Recipe 🍳

1 🗜️ Quantize the Model

Compress the ENTIRE model to 4-bit NF4 precision. A 70B model that needed 140GB now fits in 35GB!

2 📝 Add LoRA Adapters

Now add tiny LoRA layers on top of the quantized model. Only these small adapters get trained!

3 🚀 Train & Merge

Train the LoRA adapters. The frozen 4-bit model provides context; LoRA provides new knowledge!

🤯 WOW STAT: Guanaco-65B (the first QLoRA model) beat ChatGPT-3.5 on 9 out of 11 benchmarks — trained on ONE 48GB GPU in just 24 hours!

QLORA'S SECRETS

QLoRA's 3 Special Powers

01 4-bit NF4 Quantization

Use 4 bits instead of 32!

- Normal: each weight = 32 bits (full precision)
- NF4: each weight = only 4 bits (8× smaller!)
- NF = 'Normal Float' — specially designed to preserve AI weight distributions
- A 70B model: 280 GB → only 35 GB!

💡 Like describing a color with 4 crayons instead of a full 32-color box — surprisingly accurate!

02 Double Quantization (DQ)

Compress the compression!

- Quantization needs 'constants' to work properly
- DQ quantizes THOSE constants too!
- Saves an extra ~0.5 bits per parameter
- Extra ~3 GB saved on a 65B model — for free!

💡 Like zipping a zip file — you compress the thing you used to compress! Every bit counts!

03 Paged Optimizers

Borrow memory from hard drive!

- GPU memory (VRAM) is tiny & precious
- If VRAM runs out, optimizer states move to CPU RAM
- No crashes! Training continues smoothly
- Uses NVIDIA unified memory tech

💡 Like overflow parking — when the garage is full, spill into the street temporarily!

QLoRA in the Wild: Real Models 2023 – 2026

2023

Guanaco-65B
BEAT ChatGPT! Trained in 24hrs on ONE 48GB GPU!

OpenHermes 2.5
Mistral 7B + QLoRA — exceptional instruction following

NousHermes-2
Llama-2 70B QLoRA — great at complex reasoning

2024

WizardLM-2 (Microsoft)
QLoRA on Mistral 8×22B — rival to GPT-4!

Hermes-3 Llama 3.1
Llama-3.1 70B + QLoRA — top open-source model

Phi-3 Mini + QLoRA
3.8B model fine-tuned on ONE gaming GPU (8GB!)

Gemma-2 9B QLoRA
Google model fine-tuned on single consumer GPU!

Llama 3.2 3B QLoRA
Tiny model, huge performance — edge device ready!

2025

DeepSeek-R1 + QLoRA
Reasoning god — fine-tuned for specific domains cheaply

Phi-4-Mini 3.8B QLoRA
Runs on 6GB VRAM with QLoRA! Laptop AI!

Gemma-3 4B QLoRA
Google — fine-tunes in 30 min on consumer hardware!

Llama 4 Scout 17B QLoRA
17B MoE model fine-tuned on a single A100 GPU

Qwen3 7B QLoRA
Alibaba's multilingual model — cheaply customizable

Tools

HuggingFace PEFT
The go-to library for LoRA & QLoRA in Python

Unsloth 🦥
2× faster training, 80% less memory! Free & open source

LlamaFactory
Easy web UI for fine-tuning any model with QLoRA

Axolotl
Production-grade fine-tuning config tool for QLoRA

Ollama + GGUF
Run QLoRA-fine-tuned models locally on any laptop!

The Big Showdown: Full Training vs LoRA vs QLoRA

What We Compare	Full Fine-Tuning	LoRA	QLoRA
GPU Memory (70B model)	160+ GB (2× A100!)	40–80 GB (1–2 GPUs)	16–35 GB (1 GPU!) ✅
Parameters Trained	All 70B (100%)	~0.1–1% (70M–700M)	~0.1–1% (quantized) ✅
Training Cost (70B)	$50,000+ 😱	$500–5,000 😅	$50–500 🎉
Quality vs Full Train	100% (baseline)	96–99% ✅	92–97% ✅
Training Time (70B)	Weeks (multi-GPU)	Hours to days	Hours (single GPU) ✅
Who can use this?	Only big research labs	Teams with GPUs	EVERYONE! 🎓
Best tools (2025)	DeepSpeed, Megatron	HF PEFT, Unsloth	Unsloth, LlamaFactory

Remember This! 🎉

📝 LoRA = Tiny Sticky Notes for AI

Freeze the big brain, add tiny adapter layers. Only ~0.1% of parameters are trained — but the AI learns new skills perfectly!

⚡ Fine-tune Llama-3 70B with just 100M trainable params!

🗜️ QLoRA = Sticky Notes on a Compressed Brain

First squish the model to 4-bit NF4 (8× smaller). THEN add LoRA adapters. Result: fine-tune 70B models on ONE gaming GPU!

⚡ Phi-4-Mini fine-tunes on 6GB VRAM — cheaper than Netflix!

💰 Massive Cost Savings

Full GPT-3 training: $4.6 million. LoRA on Llama-3 7B: ~$10. QLoRA on a 70B model: ~$50-100. Students can now fine-tune giant models!

⚡ Alpaca-LoRA 2023: $100 → ChatGPT-level chatbot!

🌍 Used Everywhere in 2025

Llama 4, DeepSeek-R1, Gemma 3, Phi-4, Qwen3 — ALL have huge LoRA/QLoRA fine-tune ecosystems. Flux.1 LoRAs changed AI art forever!

⚡ CivitAI: 100,000+ LoRA image models — all FREE to download!